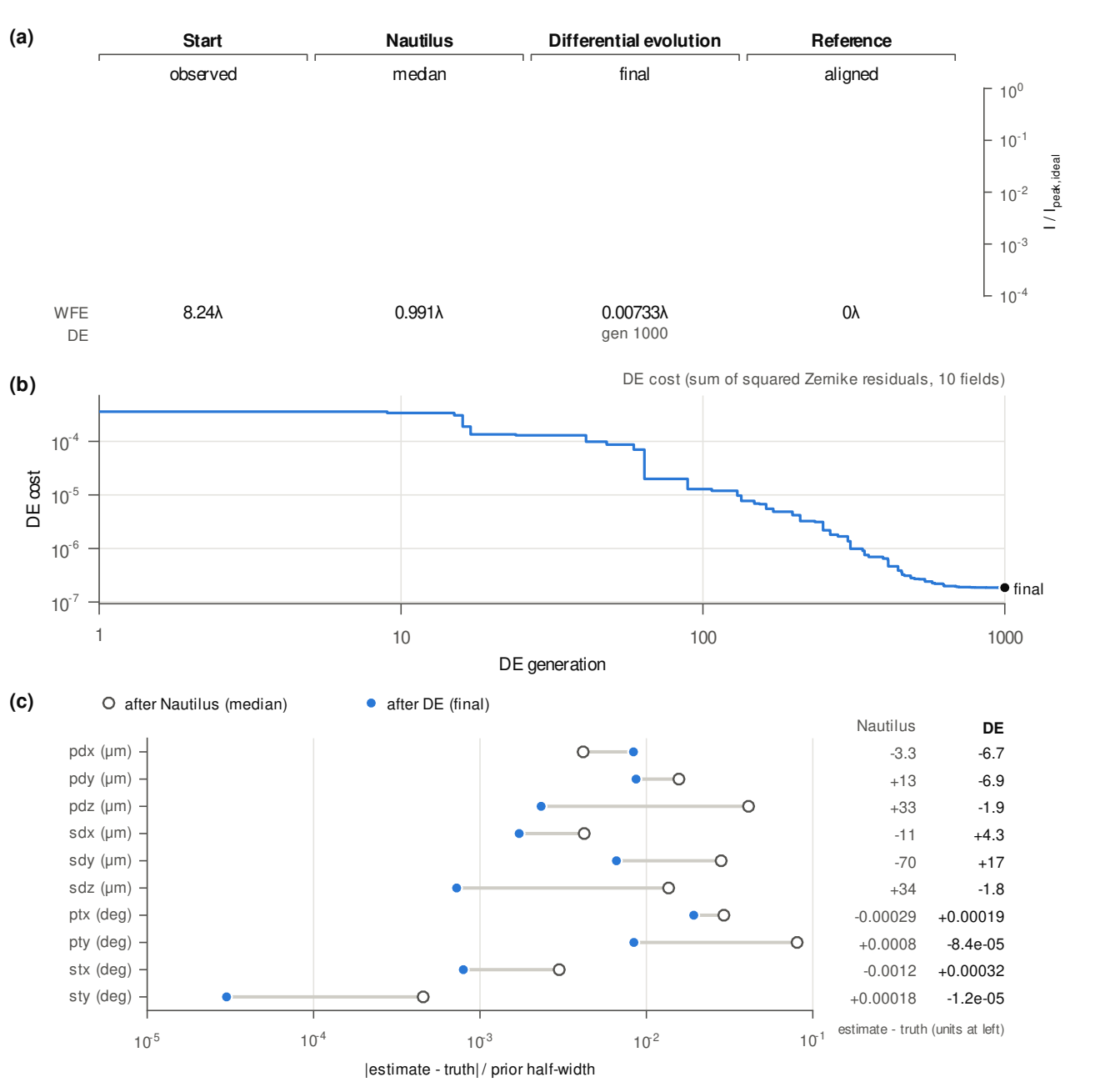
(a)
Start
Nautilus
Differential evolution
Reference
observed
median
final
aligned
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$I / I_{peak,ideal}$
WFE
DE
8.24λ
0.991λ
0.00733λ
gen 1000
0λ
(b)
DE cost (sum of squared Zernike residuals, 10 fields)
DE cost
10^{-4}
10^{-5}
10^{-6}
10^{-7}
final
1
10
100
1000
DE generation
(c)
after Nautilus (median)
after DE (final)
Nautilus
DE
pdx (μm) -3.3 -6.7
pdy (μm) +13 -6.9
pdz (μm) +33 -1.9
sdx (μm) -11 +4.3
sdy (μm) -70 +17
sdz (μm) +34 -1.8
ptx (deg) -0.00029 +0.00019
pty (deg) +0.0008 -8.4e-05
stx (deg) -0.0012 +0.00032
sty (deg) +0.00018 -1.2e-05
estimate - truth (units at left)
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
|estimate - truth| / prior half-width